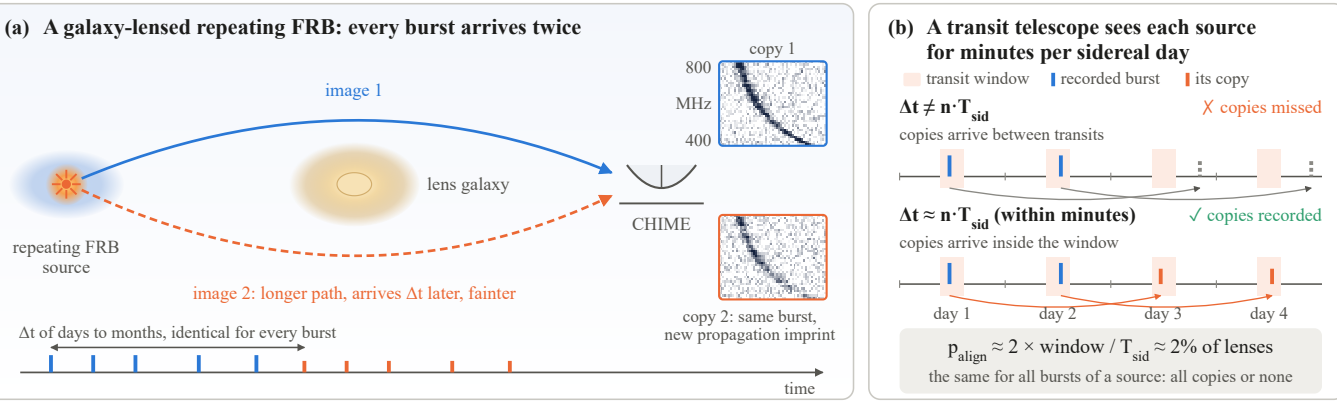

(a) A galaxy-lensed repeating FRB: every burst arrives twice
copy 1
800
MHz
400
image 1
lens galaxy
CHIME
repeating FRB
source
image 2: longer path, arrives Δt later, fainter
copy 2: same burst,
new propagation imprint
Δt of days to months, identical for every burst
time
(b) A transit telescope sees each source
for minutes per sidereal day
transit window
recorded burst
its copy
Δt ≠ n·T_sid
✗ copies missed
copies arrive between transits
Δt ≈ n·T_sid (within minutes)
✓ copies recorded
copies arrive inside the window
day 1
day 2
day 3
day 4
p_align ≈ 2 × window / T_sid ≈ 2% of lenses
the same for all bursts of a source: all copies or none